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Efficient dissolution of lignin in novel ternary deep eutectic solvents and its application in polyurethane

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ABSTRACT

Novel ternary deep eutectic solvents (DESs) were designed to efficiently dissolve the lignin for synthesizing the lignin-containing rigid polyurethane foam (LRPUF). These ternary DESs were constituted by choline chloride (ChCl), two hydrogen bond donors – glycerol (Gly) and polyethylene glycol (PEG-400). The maximum solubility of lignin in the sample DES-2 (ChCl:Gly:PEG-400 = 1:2:2) could up to 66.70 g/100 g solvent at 60 °C due to the hydrogen bonds are the main driving force, which was confirmed by the largest hydrogen bond basicity parameter β value and disappearance of crystalline peak. The main structure of the lignin did not change after dissolution in DES-2. The content of phenolic and total hydroxyl groups of the regenerated lignin increased, with reduction in their average molecular weights and polydispersity, indicating that the dissolution and regeneration of lignin in these ternary DESs can make it more homogeneous. The regenerated lignin was partially replacing for polyols in the LRPUF synthesis, the lignin replacement rate could up to 58.6% with good mechanical properties. The findings add to our understanding for efficient dissolution of lignin in DESs and for synthesizing the highly LRPUF.

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1. Introduction

Lignin, as the primary aromatic component of lignocellulosic biomass, is an abundant renewable resource. The heterogeneous structure of lignin was formed in a radical polymerization process of monolignols (coniferyl alcohol, paracoumaryl alcohol and sinapyl alcohol). Lignin is a by-product of the paper industry and a low-cost biomass waste, which has been widely regarded as an ideal raw material for the synthesis of bio-based materials and products, such as biofoam [1,2], carbon fiber [3,4], carbon quantum dots [5], anode [6], adsorbent [7], thermoplastic elastomer [8], pharmaceutical tablets [9], fertilizer [10], and so on. Polyurethane is a type of polymer with urethane structural unit (-NH-(C=O)-O-), which mainly formed by the reaction of polyisocyanate and polyol. In the literatures, there are a large number of studies on the synthesis of LRPUF. For instance, Pan et al. reported that rigid polyurethane foam was produced with 19–30% lignin addition [1]. Hayati et al. investigated that the polyol with 5% lignin can improve the compressive strength of polyurethane foam [11]. Zhang et al. reported that LRPUF with 20–50% alkylated lignin has the advantages of low density and high compressive strength [12]. Our previous study showed that the polyol was replaced with 8–37% lignin to prepare the LRPUF reinforced with pulp fiber [2]. However, the common problem exposed by the above studies is that lignin has low reactivity and is easily agglomerated to cause phase separation.

In fact, the efficient utilization of lignin is quite difficult, which offer low solubility in common organic solvents and insolubility in water [13]. Although liquefaction of lignin have been employed to break lignin down to lignin-derived intermediates into useful chemicals or fuels, such as catalytic, thermal, and biological approaches [14–16], which is still at a relatively early stage and presents significant technical challenges [17–20]. From technoeconomic standpoint, efficient solubilization and fractionation of lignin is highly desirable [21,22]. Until now, various solvent systems have been explored for lignin dissolution, including organic solvent [23,24] and ionic liquids (ILs) [25–28]. However, organic solvents are relatively corrosive and toxic, and they are likely to cause environmental pollution, which limits their industrial application. The mentioned ILs typically have following disadvantages, poor biodegradability, high price, questionable biocompatibility, high viscosity and sustainability issues, which make their large scale use not appealing at the moment [29–32]. Therefore, the development of biodegradability, low cost, low energy consuming and environment friendly solvent systems are still necessary.

DESs are obtained by mixing a hydrogen bond acceptor (HBA) and a hydrogen bond donor (HBD), these mixtures may form a eutectic system with a melting point of the mixture lower than the individual components [33,34]. Recently, DESs have been developed for dissolving lignin due to their biodegradability, low cost and environment friendly.

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Francisco et al. first studied Kraft lignin solubility in DES composing of maleic acid:valine (1:3) is up to 15 g/100 g [35]. Lynam et al. reported the dissolution of Kraft lignin in DESs such as between lactic acid and betaine, proline and choline chloride with formic acid, lactic acid and acetic acid [36]. Soares et al. showed the DES of the propionic acid: urea (2:1), which is capable of dissolving approximately 22.7 g/100 g solvent at 40 °C [37]. Liu et al. designed two different DESs to dissolve lignin using lactic acid and *N*-methylthiourea as HBD, which shows that the functional groups in DESs significantly affect lignin solubility [38]. However, the solubility of lignin in these DESs is generally not high, and lignin is easily condensed without stabilization agents under acidic conditions.

High boiling point solvents such as glycerol/PEG-400 have widely used in liquefaction of lignin [39,40], in which glycerol/PEG-400 is thought to participate in lignin with internal hydrogen bonding. However, the solubilization of lignin was generally conducted at acidic conditions, high temperatures and high solvent/substrate ratios that are typically >10:1. The polyol-based DES has potential applications in many aspects of biomass processing, such as good capability for lignin dissolution and depolymerization, excellent recyclability with scale recovery, and compatibility with microorganisms [41–43]. In addition, PEG assisted ionic liquid for pretreating the lignocelluloses to produce bioethanol [44] and PEG as a HBD for dissolving of metal oxides [45] have been investigated. Therefore, sustainable PEG based DESs with low energy consuming would be an innovation for lignin dissolution.

In this work, novel ternary DESs were designed and prepared by simply an HBA (choline chloride) and two HBDs (glycerol/PEG-400) to effectively dissolve lignin under mild condition. Kamlet-Taft parameters and the rheological properties of ternary DESs were examined. The structure and properties of the lignin precipitated after DES dissolution were studied. Finally, the performance of regenerated lignin in replacing the polyols for highly lignin-containing polyurethane foam was evaluated and discussed on the basis of its hydroxyl groups content and mechanical properties.

2. Materials and methods

2.1. Materials

Enzymatic hydrolysis lignin (EHL) in this study was separated from cellulolytic enzyme hydrolysis of corncob and supplied by the Longlive Biological Technology Co. Ltd. (Shandong, China). The chemical characteristics of the lignin were formerly reported. The raw lignin contained 88.5% Klason lignin and 4.7% acid-soluble lignin [2]. Milled wood lignin (MWL) was obtained from corncob and the processed conditions were according to the method of Bu et al. [46]. Choline chloride (ChCl, 99%), glycerol (Gly, 99%), polyethylene glycol 400 (PEG-400), 4-nitroaniline (≥ 99%), Nile red (≥ 95%), N, N-diethyl-4-nitroaniline (> 98%), 4,4'-Methylenebis(phenyl isocyanate) (MDI, 98%), and dibutyl tin dilaurate (DBTDL, 95%) were purchased from Aladdin®. Polyurethane foam stabilizer (AK 8805) was supplied by Maysta Chemical Co., Ltd. All the chemicals and reagents were utilized directly without further purification.

2.2. Synthesis of ternary DESs

PEG-400 of different molar ratio (based on the molar ratio of ChCl) was added to the mixture of ChCl and Gly (1:2, molar ratio) at room temperature. Then, the systems were heated to 60 °C for 2 h with magnetic stirring. After that, homogeneous solvents were formed and named as DES-1 (ChCl:Gly:PEG-400 = 1:2:1), DES-2 (ChCl:Gly:PEG-400 = 1:2:2), DES-3 (ChCl:Gly:PEG-400 = 1:2:3) and DES-4 (ChCl:Gly:PEG-400 = 1:2:4). As a control, the mixture of Gly and PEG-400 (1:1, molar ratio) was processed under the same conditions and named as PG.

2.3. Kamlet-Taft parameters

The dyes used for Kamlet-Taft parameters (K-T parameters) were dissolved in methanol firstly, and the concentration of 1.0×10^{-3} mol·L^{-1}. 50 μL dye solution was added into the centrifuge tube and carefully removed the residual methanol by vacuum drying at 40 °C for 0.5 h. After the residual methanol was evaporated, 2 g of the prepared solutions was mixed with the dye for testing. The mixture of dye and solutions was put into quartz cells with 1.0 cm light path length. The absorption spectra were recorded with a Shimadzu UV-1800 spectrophotometer at room temperature. The detailed procedures and equations for obtaining the K-T parameters were giving in Supporting information.

2.4. Lignin solubility determination

Vials containing 1 g of ternary DESs were placed in an oil bath. When the temperature reaches the measured temperature, consecutive additions about 0.01 g of solute were done under vigorous stirring to ensure good contact between phases. The mixture was kept at an appropriate temperature and stirred under N_2 gas. Additional EHL about 0.01 g each time was added into the mixture one by one until EHL could not be dissolved in the solvent within 2 h. For each situation, the measurement temperature was from 40 to 70 °C increased by 10 °C.

2.5. Preparation of lignin dissolution systems

The EHL and above ternary DESs were continuously stirred at 60 °C for 2 h with a solid-liquid ratio of 1:5, designated as DES-1-L, DES-2-L, DES-3-L and DES-4-L, respectively. The lignin was then precipitated with water as a reverse phase reagent and lyophilized. The regenerated lignin from above mixtures was named as L1, L2, L3 and L4, respectively. The PG was used as same condition to dissolve lignin. The dissolution system and regenerated lignin named as PG-L and L-PG, respectively.

2.6. Reusage of ternary DESs

After the lignin was regenerated and filtered, the water in the DES-2 was removed under vacuum rotary evaporation, and the recycled DES-2 was obtained. The EHL was dissolved again in recovered DES-2 at 60 °C for 2 h with a solid-liquid ratio of 1:5. The lignin was precipitated by adding appropriate amount of acid water. The above steps were repeated to obtain different recycled DES-2. The lignin solubility in different recycled DES-2 was measured respectively.

2.7. Synthesis of LRPUF

A literature protocol was followed for one-shot method LRPUF preparation [2]. The formulation of LRPUF was listed in Table S1 of the Supporting information. The [NCO]/[OH] ratio of the mixtures was set as 1.20. First mix the polyol with additives (lignin, catalyst, stabilizer and blowing agent) at ambient temperature for 5 min, and then thoroughly mix all ingredients using a high speed mixer (400 rpm) to disperse different amounts of lignin in the polyol, and subsequently, predetermined MDI was added into paper cup and stirred vigorously with a high speed mixer (2000 rpm) for 30 s. Then, the resulting mixture was quickly poured into a cubic paper container at room temperature to obtain LRPUF. Finally, the obtained LRPUF were aged in a vacuum drying oven for 24 h.

2.8. Characterization of ternary DESs, regenerated lignin and LRPUF

Fourier transform infrared (FT-IR) were scanned using KBr pellets and recorded by Bruker Vertex 70 under a resolution of 4 cm^{-1} from 400 to 4000 cm^{-1}. The FT-IR spectra of the LRPU were used the ATR mode for testing. Differential scanning calorimetry (DSC) was measured

by DSC-Q2000 thermal analyzer (TA Instruments, USA). The measurement conditions were performed in the temperature from −80 to 100 °C with the heating rate of 10 °C/min. The thermogravimetric analysis (TGA) were performed using a NETZSCH STA449F3 (NETZSCH Scientific Instruments, Germany) under a N_2 atmosphere with the heating rate of 10 °C/min. Rheological properties were performed by rheometer (DHR-3, TA Instruments, USA). The test conditions were measured in the temperature from 25 to 80 °C with the frequency range of 150 rad/s, using 20 mm Peltier Plate with the gap of 53 μm. Frequency sweep tests were performed inside the linear viscoelastic regime, in a shear rate of 0.1–300 s^{-1}. 2D-HSQC NMR (Two-dimensional heteronuclear single quantum coherence NMR) were recorded on a Bruker AV-III 400 MHz spectrometer (Germany) at 25 °C in DMSO-d_6 as the solvent. The spectral widths were 5000 and 20,000 Hz for the ^1H and ^{13}C dimensions, respectively. The central solvent (DMSO) peak was used as an internal chemical shift reference point (δ_C/δ_H 39.5/2.49). The content of phenolic hydroxyl groups were verified by Folin-Ciocalteu method [47]. The molecular weight distribution of the lignin samples was examined by gel permeation chromatograph (GPC, Agilent 1200, USA) with a PL-gel 10 μm mixed-B 7.5 mm ID column. The lignin samples were dissolved in tetrahydrofuran (2 mg/mL) and 20 μL samples were injected with a flow rate of 0.5 mL/min. The morphology of sample LRPUF was examined using a field emission scanning electron microscope (SEM, S4800, Hitachi, Japan). The compressive strengths of foam were measured at ambient temperature using a universal testing machine (AI-7000-NGD, Gotech Testing Machines Inc., China). The size of the specimen was 30 mm × 30 mm × 30 mm (length × width × thickness) at the cross-head speed of 10 mm/min. Compressive stress is performed at a 10% strain parallel to the rising direction of the foam, according to GB/T 8813-2008. For each compression test, three replicate samples were tested and the average and standard deviation were taken.

3. Results and discussion

3.1. Solubility of EHL

The solubility of EHL in PG and different ternary DESs at various temperatures is given in Table 1. The solubility of EHL in PG is linearly related to temperature. With increasing temperature from 40 to 60 °C, the solubility of EHL in ternary DESs increased dramatically, and then reached the maximum value at 60 °C. Whereas, the lignin solubility was measured at 80 °C in DES-2, which is 66.65 g/100 g. The result shows that the lignin solubility tends to be constant above 60 °C. It is worth noting that the sample DES-2 shows maximum solubility of EHL with 66.70 g/100 g solvent at 60 °C. Compared with the published literatures based on choline chloride as HBA (Table 2), the largest solubility of EHL was achieved in DES-2 at 60 °C.

β value in the Kamlet−Taft hydrogen bond basicity parameter presents the ability of the solvent system to donate electrons to form a hydrogen bond with protons from solute [49,50]. The relationship between β value and maximum lignin solubility in different solvents at room temperature was shown in Fig. 1. It is found that the maximum β value corresponding to the maximum solubility of lignin at 60 °C in DES-2. The K-T parameters of PG and ternary DESs are given in Table S2. The results show that lignin has better dissolving ability in ternary DESs with a large subtraction value of α and β. When the subtraction value of α and β is large, the excess HBA and HBD in the solvent will form the hydrogen bonds between the lignin and ternary DESs, which will promote the dissolution of lignin. In addition, the K-T parameters can also analyze the influence of solvent polarity on the lignin dissolving ability. What stands out in this table is the similar α and β value between DES-2 and DES-3 which shows the larger π* value, the better lignin solubility.

Table 1
Maximum solubility of EHL in ternary deep eutectic solvents (g/100 g solvent).

Entry	Temperature			
	40 °C	50 °C	60 °C	70 °C
PG	38.40	41.47	44.66	54.71
DES-1	44.23	50.10	62.96	56.04
DES-2	50.05	58.13	66.70	66.61
DES-3	52.25	55.73	66.07	63.82
DES-4	55.85	58.36	66.04	59.30

The solvent viscosity is another significant factor affecting the lignin solubility in the solvent systems. The viscosity of ternary DESs is related to the nature of its components and temperature, which is primarily determined by hydrogen bonding, van der Waals forces and electrostatic interactions [51,52]. The viscosity depending on the temperature for PG and ternary DESs is depicted in Fig. 2. Obviously, the viscosity of DES-2 is the lowest of these samples. As the temperature increases, the viscosity of all samples decreased. The viscosity of DES-2 is minimum in all ternary DESs. Low viscosity does not hinder the diffusion of lignin molecules in solvents, which also related to the maximum solubility of lignin in DES-2 at 60 °C. In order to explore the influence of different solubility of EHL on the viscosity of the system, the viscosity of the DES-2 with different solubility of EHL at 60 °C was measured and shown in Fig. S1. It is obvious that the viscosity of the dissolution system increases as the amount of EHL dissolved increases. This is caused by the formation of entanglement of lignin molecules in the solution, resulting in strong concentration dependence.

The viscosity of PG, PG-L, DES-2 and DES-2-L in different shear rate is given in Fig. S2. Generally, the viscosity of pure PG and DES-2 has hardly changed in different shear rate. After the dissolution of EHL, the shear thinning behavior was observed in samples PG-L and DES-2-L, which is attributed to the breakage of the interconnected PEG and Gly domains separated by the lignin phase [53]. This phenomenon is due to the presence of hydrogen bonds in the dissolution systems. Compared with pure solvents, the viscosity of samples PG-L and DES-2-L increases rapidly after the dissolution of lignin. This due to the dissolution of lignin broke the hydrogen bond in solvents and newly formed the hydrogen bond between EHL and solvents, which promoting the dissolution of EHL in the DES-2 [39].

FT-IR spectra of the interaction between EHL and solvents are shown in Fig. S3. The peaks at 3200–3600, 2900, 1647 and 1253 cm^{-1} are assigned to the O—H, C—H stretching, C═O stretching and C—O stretching vibration [54]. The peak of C═O at 1647 cm^{-1} in EHL slightly shift to 1703 cm^{-1} in the samples of PG-L and DES-2-L, indicating that the hydrogen bond probably formed in EHL with PG and DES-2. The dissolution of EHL also affects the melting and crystallization behavior of PG and DES. The DSC results of PG, PG-L, DES-2 and DES-2-L are exhibited in Fig. 3, it show the obvious peaks of melting and crystallization due to the ordered crystalline structure of PEG/Gly formed by abundant hydrogen bonds in the samples of PG and DES-2 [55,56]. In contrast, the peaks of melting and crystallization disappeared after the dissolution of EHL to DES-2. These substantive changes can be ascribed to the presence of EHL leading to the disordered arrangement of DES-2, where the hydrogen bond network of DES-2 is affected by the newly formed hydrogen bond between DES-2 and EHL [57,58]. Meanwhile, the disappearance of melting peak also suggests that the sample DES-2 interacts favorably with EHL and constitutes a miscible blend [59].

The most important considerations for DES dissolving lignin are economy and solvent recyclability, and the recovered solvent without loses its dissolution efficiency. The recovered mass of the ternary DES-2 after recycling was found to be more than 98%, which indicated that ternary DES-2 as a potentially good recyclable solvent. The lignin solubility in in pure and recycled DES-2 was shown in Fig. S4. The lignin solubility can be maintained at 61.3 g/100 g, 61.0 g/100 g and 58.7 g/100 g for recycling cycles 1, 2 and 3 respectively.

Table 2
Dissolution of lignin in DESs with ChCl as HBA.

DES	Lignin solubility (g/100 g)	Conditions	Reference
ChCl:lactic acid (1:5)	7.77	60 °C	Francisco et al. [35]
ChCl:formic acid (1:2)	14	60 °C	Lynam et al. [36]
ChCl:phenol (2:1)	67.87	90 °C, ultrasonic	Malaeke et al. [48]
ChCl:α-naphthol (1:1)	31.42	90 °C, ultrasonic	Malaeke et al. [48]
ChCl:*N*-methylthiourea (1:2)	59.49	70 °C	Liu et al. [38]
ChCl:Gly:PEG-400 (1:2:2)	66.70	60 °C	This work

3.2. Characterization of EHL and regenerated lignin

FT-IR spectra of raw EHL and regenerated lignin are shown in Fig. S5. The peaks at 3491, 2933, 1703 and 1215 cm^{-1} are attributed to the O—H stretching, C—H stretching, C═O stretching and C—O stretching, respectively. The bands at 1600 and 1508 cm^{-1} are ascribed to the skeleton vibration of the benzene ring in lignin. It is clear that the chemical structure of the regenerated lignin did not change after dissolution in different solvents. The relative intensity of the band at 3100–3600 cm^{-1} attributed to O—H stretching increased greatly. In order to use regenerated lignin well in the synthesis reaction, the hydroxyl value of regenerated lignin was determined.

The content of phenolic hydroxyl groups and total hydroxyl groups in the regenerated lignin, as shown in Fig. S6, including the effects of different DESs, temperature and dissolution time. It was found that the phenolic and total hydroxyl content of L2 were the highest at 60 °C for 1 h, reaching 2.640 mmol/g and 6.826 mmol/g, respectively. Furthermore, the effect of different temperatures on the content of phenolic and total hydroxyl groups of lignin regenerated from DES-2 was explored. As the reaction temperature rises, the phenolic hydroxyl content tends to be stable at 60 °C, while the total hydroxyl content is up to 6.950 mmol/g at 70 °C. Finally, the effect of different dissolution time on the content of phenolic and total hydroxyl groups of regenerated lignin in the DES-2 system at 60 °C was detected. It was found that the phenolic and total hydroxyl content of the regenerated lignin at 2 h were up to 3.360 mmol/g and 7.976 mmol/g, respectively.

In this study, to elaborate the structural features of lignin after dissolution in DES-2, 2D-HSQC NMR spectra of the raw EHL and L2, MWL and MWL-DES-2 (regenerated after MWL dissolution in DES-2 at 60 °C for 2 h) are shown in Fig. 4, and the relevant signals are shown in the Table S3. Fig. 5 shows the main classical structures in the MWL and MWL-DES-2 according to Fig. 4. In the side chain region (δ_C/δ_H 50–90/2.5–6.0), the presence of *β-O*-4 substructure (A) was detected by cross peaks at δ_C/δ_H 59.8/3.59 (A_γ), 63.8/4.36 (A'_γ), 71.6/4.85 (A_α), and 85.9/4.10 (A_β). The proportion of *β-O*-4 in EHL and L2 is 20.21% and 19.49%, whereas the proportion of *β-O*-4 in MWL and MWL-DES-2 is 54.03% and 53.16%, respectively. In the aromatic region (δ_C/δ_H 90–160/6.0–8.0), the signal peaks of the S (103.8/6.71), G (110.9/6.98, 115.0/6.68 and 119.1/6.64) and H (121.9/7.12 and 128.0/7.05) can be detected. The presence of ferulic acid ester (FA) and *p*-coartate (*p*CA) signals can also be observed. The S/G ratio in EHL and L2 is 3.54 and 3.57, whereas the ratio of S/G in MWL and MWL-DES-2 is 0.31 and 0.29, respectively. These results indicate that the structure of lignin did not change after dissolution in DES-2.

The molecular weight of lignin is an important influence on lignin properties. The molecular weight and PDI of EHL and regenerated lignin are given in Table 3. Compared with those of EHL, the $\overline{M}_w$ and PDI of all regenerated lignin decreased, which is because EHL could destroy the ordered structure of ternary DESs through hydrogen bonding [60]. Meanwhile, the hydrogen bonds between lignin molecules are also broken, and lignin is no longer easy to aggregate, which making lignin molecular weight reduced and molecular weight distribution more homogeneous. As expected, the $\overline{M}_w$ and PDI of sample L2 were lower than other samples. These results suggested that more homogeneous lignin fractions could be obtained in DES-2. The thermal stability of EHL and regenerated lignin was also performed, their different thermal degradation curves and data are given in Fig. S7 and Table S4. The overall thermal decomposition behavior of raw EHL and L2 was similar. Compared with those of EHL, the $T_{10\%}$ mass loss and R_{780} of sample L2 decreased, indicating that the thermal stability of regenerated lignin after dissolving EHL in DES-2 decreased, which was probably due to the decrease in molecular weight of EHL after dissolution in DES-2.

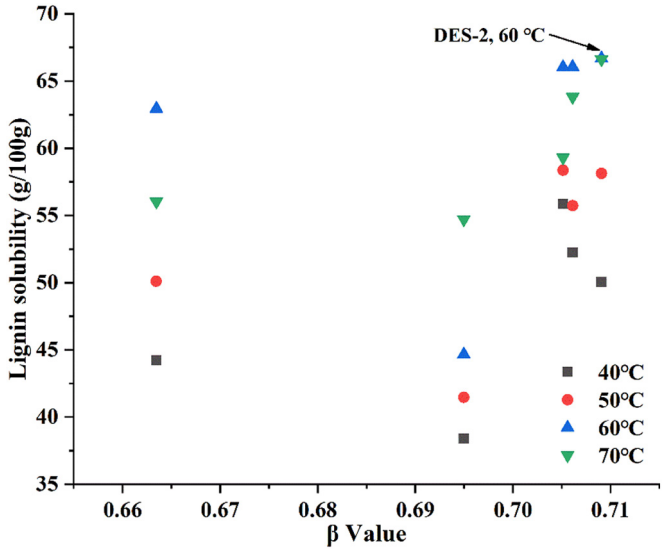


Fig. 1. The relationship between the β value and lignin solubility.

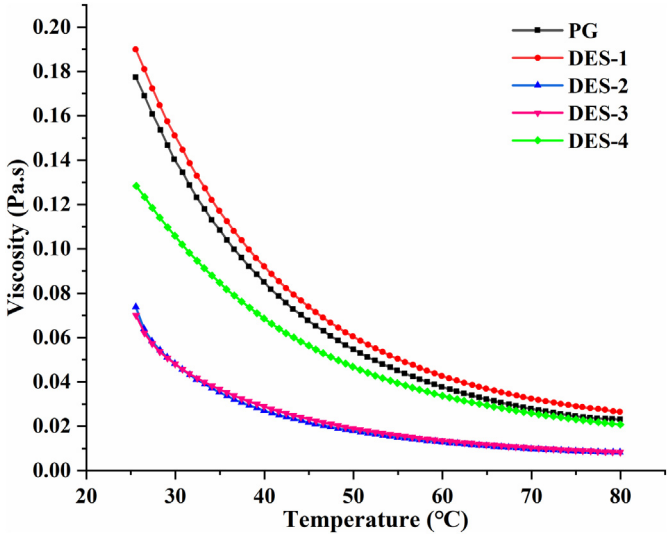


Fig. 2. Viscosity of PG and ternary DESs in different temperature.

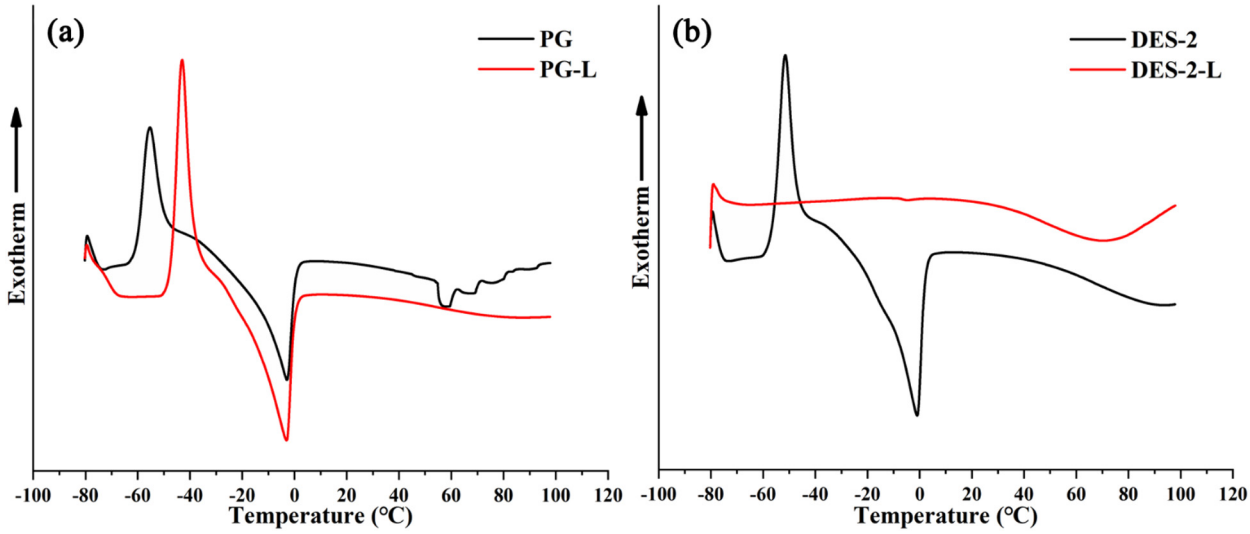


Fig. 3. DSC results of samples: (a) PG and PG-L; (b) DES-2 and DES-2-L.

Based on the above results of this study, an understanding of interaction of lignin and ternary DESs can be presented (Scheme 1).

3.3. Synthesis and characterization of LRPUF

The sample L2 was used to partially replace the polyol to prepare polyurethane foam. The result showed that the highest lignin replacement rate can up to 58.6%, which still meet the national standard GB/T 29288-2012 for rigid polyurethane foam. To the best of our knowledge this is the maximum amount of lignin replacement rate which reported for synthesizing lignin-containing polyurethane foam. At the same time, the micro-morphology of the cell structure of LRPUF was shown in Fig. 6. No aggregated particles were observed for all LRPUF, showing good interfacial compatibility between regenerated lignin and PU matrix. When the lignin replacement rate is 10–30%, the cell structure of the foam is uniform. However, the lignin replacement rate reached 40%, the cells began to rupture. And a large number of cells were broken at the lignin replacement rate reached 50%.

The compressive stress-strain curves of LRPUF are shown in Fig. 7(a). Compared with those of control sample, LRPUF present the lower elastic modulus and higher compressive stress. At 10% compressive strain, the maximum compressive stress of the sample LRPUF-30 can reach 673 kPa, whereas compression strength can up to 356 kPa at a lignin replacement rate of 50%. Fig. 7(b) shows the density of LRPUF. Comparing the density of the sample LPRF-0 (22.78 kg/m^3), the density of the

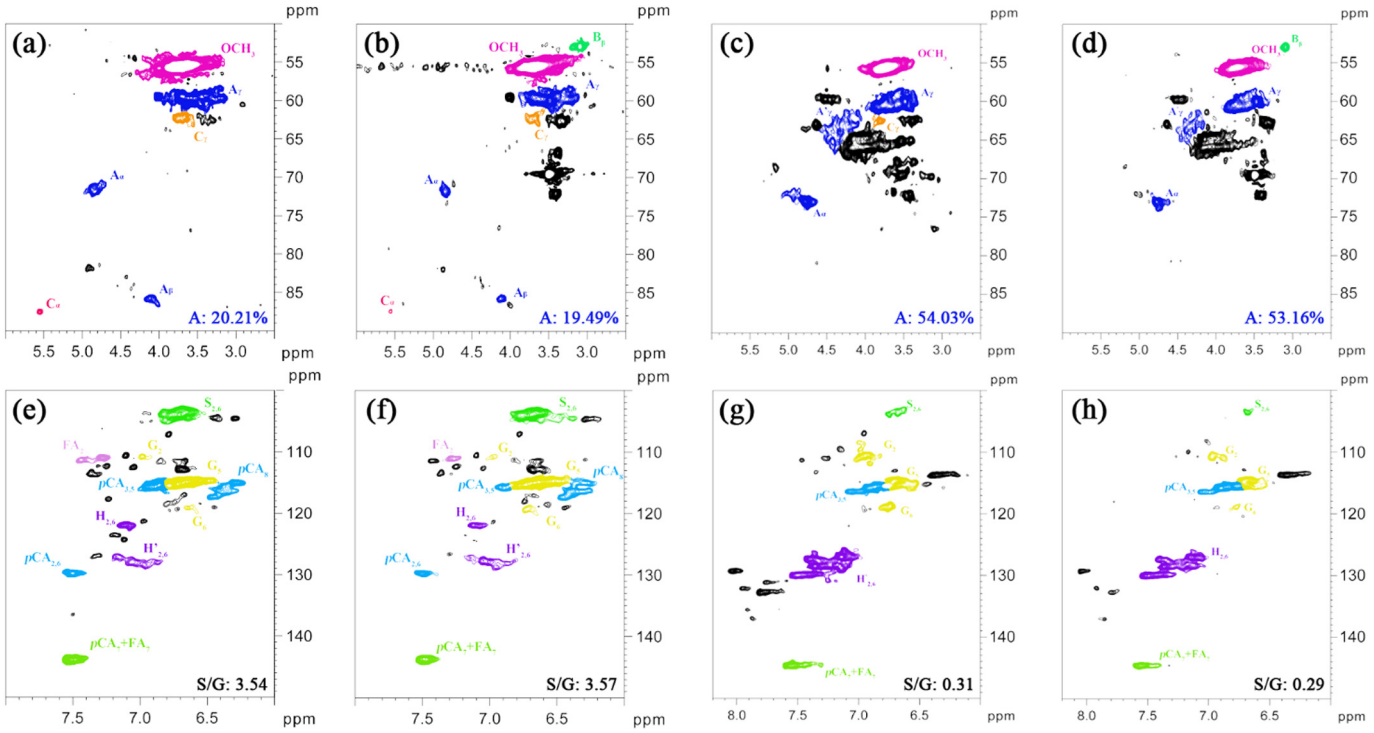


Fig. 4. Side-chain (a, b, c and d) and aromatic (e, f, g and h) regions in 2D-HSQC NMR spectra of four lignin samples (EHL: a and e; L2: b and f; MWL: c and g; MWL-DES-2: d and h).

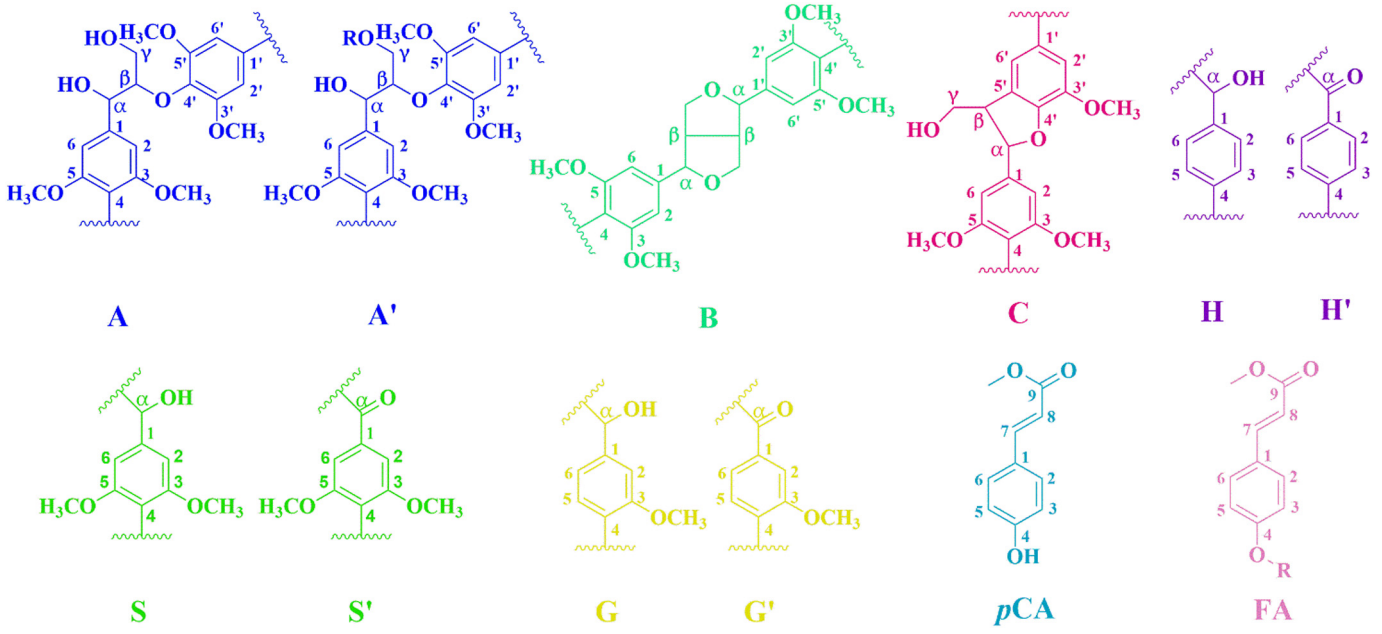


Fig. 5. Main classical structures in the lignin: (A) *β-O-4* linkages; (A') γ-acylated *β-O-4'* substructures; (B) *β-β* substructures; (C) phenylcoumarane structures formed by *β-5'/α-O-4'* linkages; (H) *p*-hydroxyphenyl units; (H') oxidized *p*-hydroxyphenyl units; (S) syringyl unit; (S') oxidized syringyl unit; (G) guaiacyl unit; (G') oxidized guaiacyl units with a C_α ketone; *p*-coumarate *p*CA, and ferulate FA; color coding is according to the structures shown in Fig. 4.

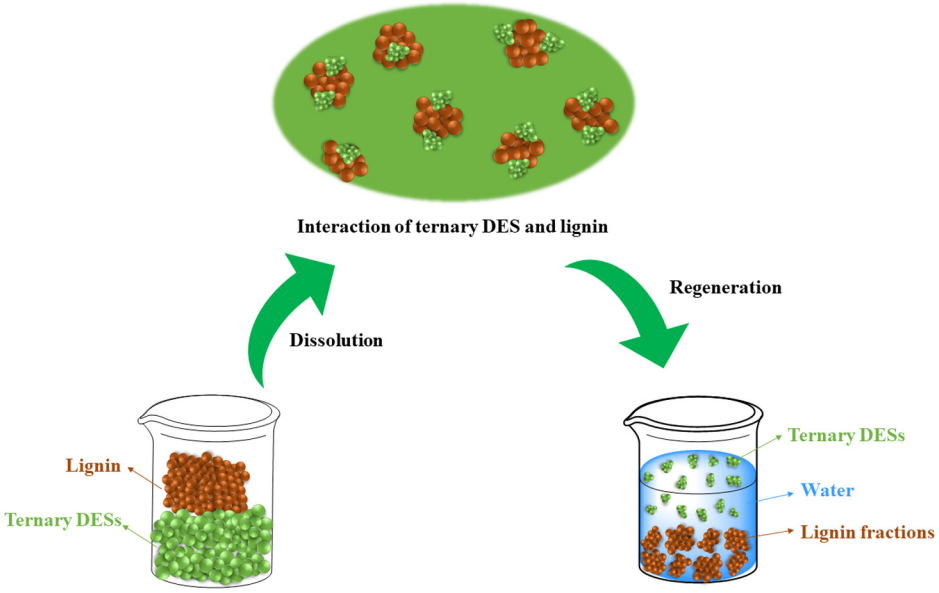
Table 3
Molecular weight and PDI of raw EHL and regenerated lignin fractions.

Entry	$\overline{M}_n$ (g·mol^{-1})	$\overline{M}_w$ (g·mol^{-1})	PDI
EHL	1530	2560	1.67
L1	1106	1507	1.36
L2	1092	1485	1.35
L3	1185	1711	1.44
L4	1150	1584	1.38

samples tends to increase, and the sample LPRF-30 has maximum density up to 64.13 kg/m^3. However, with the increasing of regenerated lignin in the polyurethane matrix, the apparent density of the samples LPRF-40 and LPRF-50 sharply decreased. The decrease in apparent density is closely related to the morphology and structure of the LRPUF, because of the lignin causes irregular and larger cells in polyurethane foam [1].

4. Conclusion

Ternary DESs were successfully synthesized under mild conditions using ChCl as HBA and, Gly/PEG-400 as HBD. The maximum solubility of EHL in sample DES-2 could reach 66.70 g/100 g solvent at 60 °C, because of the largest hydrogen bond basicity parameter β parameter and lowest viscosity. The main structure of the lignin did not change after dissolution in DES-2. The molecular weight and PDI of EHL



Scheme 1. Hydrogen bonding interaction of lignin and ternary DES.

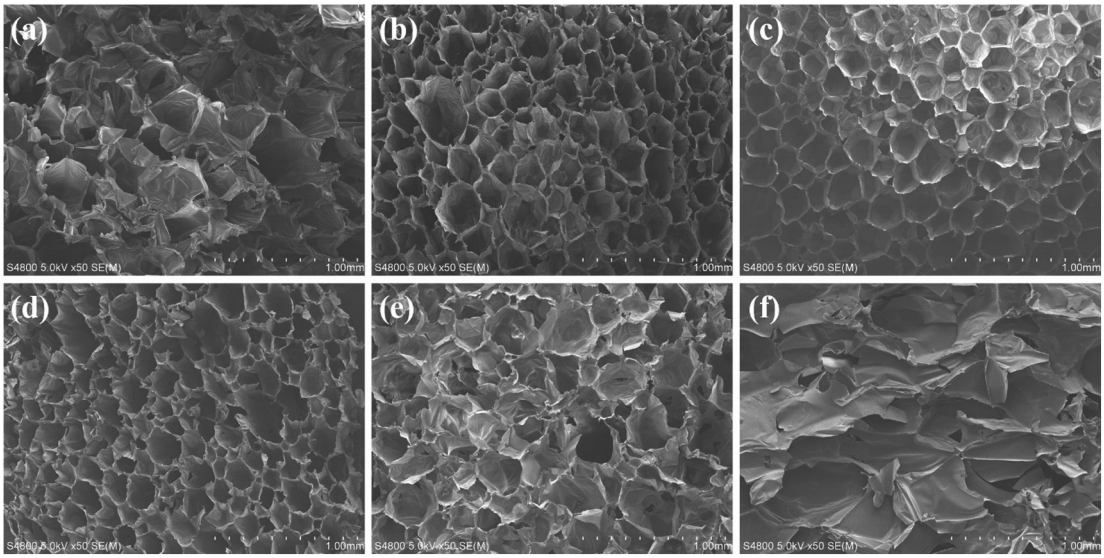


Fig. 6. SEM images (50 ×) of polyurethane foams: (a) LRPUF-0, (b) LRPUF-10, (c) LRPUF-20, (d) LRPUF-30, (e) LRPUF-40 and (f) LRPUF-50.

decreased after dissolution in DES-2. The dissolution of lignin in ternary DESs were promoted by the hydrogen bonds formation between lignin and ternary DESs. These results indicate that the dissolution and regeneration of EHL in these ternary DESs can make EHL more homogeneous. The LRPUF showed good mechanical properties and lignin replacement rate. The present study makes several noteworthy contributions to lignin dissolution in DESs and highly lignin-containing polyurethane synthesis.

Author statement

The author's individual contributions are shown as follows:

Bailiang Xue: Conceptualization, Methodology, Writing-Review & Editing, Funding acquisition.

Yang Yang: Conceptualization, Methodology, Formal analysis, Writing-Original Draft, Writing-Review & Editing, Visualization.

Rui Tang: Formal analysis, Validation, Visualization.

Danwei Xue: Validation, Visualization.

Yongchang Sun: Resources, Validation.

Xinping Li: Supervision, Funding acquisition.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ijbiomac.2020.07.153.

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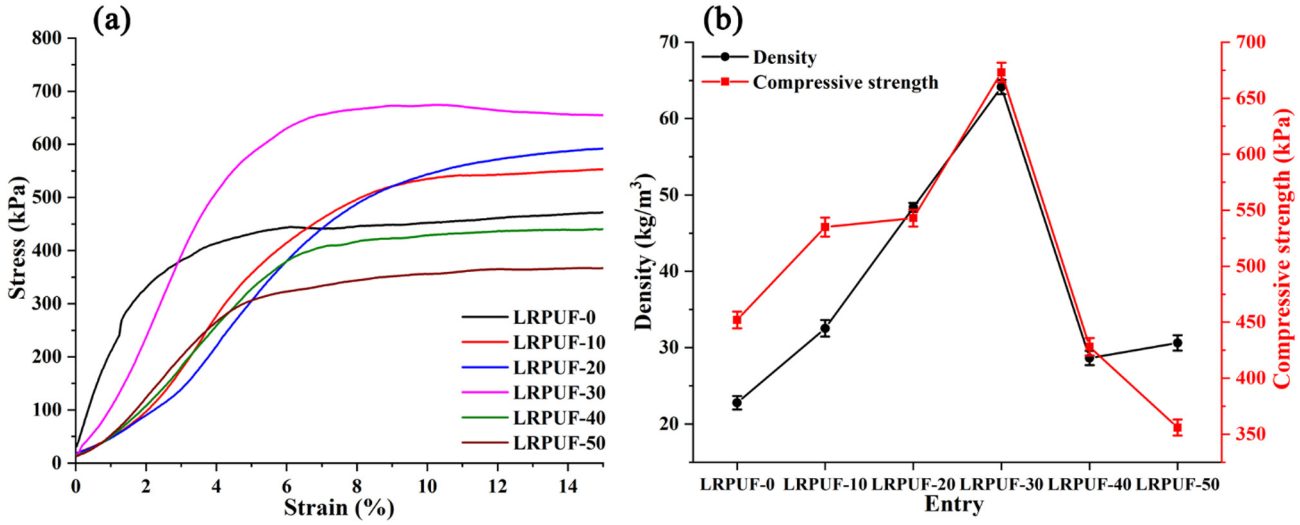


Fig. 7. (a) Compressive stress-strain of LRPUF and (b) the density and compressive strength of LRPUF.

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